NATIONAL TEACHERS COLLEGE

School of Arts, Sciences and Technology
Bachelor of Science in Information Technology

SAMPLE TEMPLATE AND ANSWER

4.4 Software Testing Plan and 4.5 User Acceptance Testing (UAT) and Evaluation

Chapter 4: Methodology, Testing, and Evaluation

How to Use This Guide

Each section below has two parts. The **TEMPLATE** part tells you what to include and gives you a blank table to fill in. The **SAMPLE ANSWER** part shows how a finished section may look.

The sample uses an imaginary system, **LibraTrack: A Web-Based Library Borrowing and Inventory System for National Teachers College**. All test cases, results, and scores are **made up for illustration only**. Replace them with the actual modules, test data, and results of your own capstone. Never copy the sample numbers into your paper.

In the sample, the main modules of LibraTrack are Login, Book Management, Borrowing, Returns and Fines, Member Management, and Reports.

4.4 Software Testing Plan

TEMPLATE: what to include

- **Purpose:** why the system was tested and what quality it aims to prove.
- **Testing approach:** the levels of testing you performed (unit, integration, system) and, in one sentence each, what they cover.
- **Test environment:** the hardware, software, browsers, and database used during testing.
- **Testers and schedule:** who tested (proponents, adviser, IT experts) and when.
- **Test case format:** the columns you use in your test tables.
- **Pass/fail criteria:** when a test case counts as passed, and when testing may stop.
- **Defect handling:** what you did when a test failed (fix, retest, record).

SAMPLE ANSWER

The proponents prepared a software testing plan to make sure that LibraTrack works as specified in the requirements analysis (Section 4.2) before it was given to the users. Testing was done in three levels: unit testing, integration testing, and system testing. The results of these tests were recorded in test case tables, and all failed cases were corrected and retested before User Acceptance Testing.

Table 4.1. Levels of Testing

Level	What Was Tested	Tested By	Schedule
Unit Testing	Individual functions and modules, such as login validation and fine computation	Proponents	Week 1 to 2 of each sprint
Integration Testing	Modules working together, such as Borrowing with Book Inventory	Proponents	End of each sprint
System Testing	The whole system, including functionality, security, performance, and compatibility	Proponents and adviser	Final development phase

Table 4.2. Test Environment

Item	Specification
Hardware	Laptop with Intel Core i5, 8 GB RAM, 256 GB SSD; Android smartphone for mobile testing
Operating System	Windows 11; Android 13
Server Software	XAMPP (Apache 2.4, PHP 8.2, MySQL 8.0)
Browsers	Google Chrome, Mozilla Firefox, Microsoft Edge (latest versions)
Testing Tools	PHPUnit for unit tests, Postman for API checks, Google Chrome DevTools for performance

Pass/fail criteria. A test case is marked **Pass** when the actual result is the same as the expected result. It is marked **Fail** when the actual result is different, when the system shows an error, or when it stops working. A failed test case was corrected by the proponents and tested again. Testing moved to the next level only when all test cases of the current level had passed.

4.4.1 Unit Testing

TEMPLATE: what to include

- Explain that unit testing checks the smallest parts of the system (functions, forms, and modules) one at a time.
- State the technique used (for example, white-box testing) and the tools (for example, PHPUnit, JUnit, or manual testing).
- List the test cases in a table. Cover **valid input**, **invalid input**, and **boundary values** (such as the lowest and highest allowed values).
- Summarize the results: number of test cases, how many passed on the first run, and what was fixed.

Table 4.x. Unit Test Cases (Blank Template)

ID	Function / Module	Test Description	Test Data / Input	Expected Result	Actual Result	Status
[UT-01]	[function name]	[what is checked]	[sample data]	[what should happen]	[what happened]	[Pass/Fail]
[UT-02]						

SAMPLE ANSWER

Unit testing was performed on the smallest parts of LibraTrack using the white-box technique. Each function was tested with valid data, invalid data, and boundary values using PHPUnit. Table 4.3 shows selected unit test cases.

Table 4.3. Unit Test Cases of LibraTrack

ID	Function / Module	Test Description	Test Data / Input	Expected Result	Actual Result	Status
UT-01	validateLogin()	Login with correct credentials	Username: staff01; correct password	User is redirected to the dashboard	User was redirected to the dashboard	Pass

ID	Function / Module	Test Description	Test Data / Input	Expected Result	Actual Result	Status
UT-02	validateLogin()	Login with wrong password	Username: staff01; wrong password	Message "Invalid username or password" is shown	Message was shown	Pass
UT-03	computeFine()	Book returned 3 days late (₱5.00 per day)	Days overdue: 3	Fine is ₱15.00	Fine was ₱15.00	Pass
UT-04	computeFine()	Book returned on the due date (boundary)	Days overdue: 0	Fine is ₱0.00	Fine was ₱0.00	Pass
UT-05	addBook()	Add a book with an existing ISBN	ISBN: 978-971-0000-01-1 (already in database)	Record is rejected with a duplicate message	Record was rejected	Pass
UT-06	searchBook()	Search with an empty keyword	Keyword: (blank)	Prompt "Please enter a keyword"	Whole book list was shown (first run); prompt shown after fix	Fail, then Pass after retest

Six unit test cases are shown. Five (83.3%) passed on the first run. UT-06 failed because the search function did not check for a blank keyword. The proponents added the validation, ran the test again, and the case passed. All unit test cases were passed before integration testing began.

4.4.2 Integration Testing

TEMPLATE: what to include

- Explain that integration testing checks whether modules that passed unit testing work correctly **together** and pass data correctly.
- State your approach (for example, incremental, bottom-up, or top-down).
- List the modules that are connected, the test scenario, the expected result, the actual result, and the status.
- Summarize the results and any defects that were fixed.

Table 4.x. Integration Test Cases (Blank Template)

ID	Modules Integrated	Test Scenario	Expected Result	Actual Result	Status
[IT-01]	[Module A + Module B]	[what is done]	[what should happen]	[what happened]	[Pass/Fail]
[IT-02]					

SAMPLE ANSWER

Integration testing was done using the incremental approach. Modules were combined one at a time at the end of each sprint, and the data passed between them was checked in the database. Table 4.4 shows the results.

Table 4.4. Integration Test Cases of LibraTrack

ID	Modules Integrated	Test Scenario	Expected Result	Actual Result	Status
IT-01	Login + Dashboard	Log in as Librarian, then as Student	Each user sees only the menu allowed for the role	Menus matched the roles	Pass

ID	Modules Integrated	Test Scenario	Expected Result	Actual Result	Status
IT-02	Borrowing + Book Management	Borrow one copy of a book that has 3 available copies	Available copies become 2	Available copies became 2	Pass
IT-03	Returns + Fines	Return a book 4 days late	Fine record of ₱20.00 is saved to the member's account	Fine record of ₱20.00 was saved	Pass
IT-04	Borrowing + Member Management	Borrow using a suspended member account	Transaction is blocked with a message	Transaction was blocked	Pass
IT-05	Reports + Database	Generate the monthly borrowing report	Report totals equal the records in the database	Totals matched	Pass

All five integration test cases (100%) passed. The results show that data moves correctly between the modules of LibraTrack.

4.4.3 System Testing

TEMPLATE: what to include

- Explain that system testing checks the **complete, integrated system** against the requirements in Section 4.2.
- Cover different types of tests: **functional**, **security**, **performance**, **compatibility**, and **reliability**.
- For performance tests, give measurable targets (for example, page load of 3 seconds or less with 30 users).
- List the test cases and results, then summarize.

Table 4.x. System Test Cases (Blank Template)

ID	Test Type	Test Scenario	Expected Result	Actual Result	Status
[ST-01]	[Functional]	[what is done]	[what should happen]	[what happened]	[Pass/Fail]
[ST-02]					

SAMPLE ANSWER

System testing was done on the complete LibraTrack system in the environment shown in Table 4.2. The system was checked against the functional and non-functional requirements listed in Section 4.2. Table 4.5 shows the results.

Table 4.5. System Test Cases of LibraTrack

ID	Test Type	Test Scenario	Expected Result	Actual Result	Status
ST-01	Functional	Complete the whole process: register member, add book, borrow, return, and print receipt	Every step is completed without error	All steps were completed	Pass
ST-02	Security	Enter ' OR '1'='1 in the login form	Login is rejected	Login was rejected	Pass

ID	Test Type	Test Scenario	Expected Result	Actual Result	Status
ST-03	Security	Student account opens the admin page by typing its URL	Access is denied	Access denied message was shown	Pass
ST-04	Performance	30 users open the book search page at the same time	Average page load of 3 seconds or less	Average load time was 2.1 seconds	Pass
ST-05	Compatibility	Open the system in Chrome, Firefox, Edge, and mobile Chrome	Pages display and work correctly in all	Pages displayed correctly in all four	Pass
ST-06	Reliability	Keep the system running for 8 hours with regular transactions	No crash or data loss	No crash or data loss occurred	Pass

All six system test cases (100%) passed. The results show that LibraTrack meets the functional requirements and the security, performance, compatibility, and reliability targets set by the proponents.

4.5 User Acceptance Testing (UAT) and Evaluation

TEMPLATE: what to include

- **Purpose:** UAT lets real users try the system and confirm that it meets their needs. Evaluation measures software quality using the **ISO/IEC 25010** standard.
- **Respondents:** who they are, how many, and how they were chosen (for example, purposive sampling of end-users and IT experts). Give a table.
- **Instrument:** the questionnaire, the ISO/IEC 25010 quality characteristics it covers, and how it was validated (checked by your adviser or an expert). Put the full questionnaire in Appendix C.
- **Rating scale:** a 5-point scale with ranges for the mean and a verbal interpretation.
- **Procedure:** how the users tried the system, and how and when they answered the questionnaire.
- **Statistical treatment:** the formula used (usually the weighted mean).
- **UAT sign-off:** a form signed by the client or users to confirm acceptance.
- The mean scores and their discussion go in **Chapter 5, Section 5.2 and 5.3**, not here.

SAMPLE ANSWER

Purpose. User Acceptance Testing (UAT) was conducted to find out whether LibraTrack meets the needs of its intended users. After the users tried the system, they evaluated its quality using a questionnaire based on the ISO/IEC 25010 software product quality standard.

Respondents. The respondents were chosen through purposive sampling, that is, those who would actually use the system or who are qualified to judge its technical quality.

Table 4.6. Respondents of the Study

Group	Number of Respondents	Percentage
Librarian and library staff	5	12.5%
Students (borrowers)	30	75.0%
IT experts	5	12.5%
Total	**40**	**100.0%**

Instrument. The questionnaire covered the eight quality characteristics of ISO/IEC 25010: Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability. Each characteristic had five statements. The questionnaire was checked by the adviser and two IT experts before use. A copy is attached as Appendix C.

Rating scale. Respondents rated each statement from 1 to 5. The mean scores were interpreted using the scale in Table 4.7.

Table 4.7. Rating Scale and Verbal Interpretation

Scale	Mean Range	Verbal Interpretation
5	4.21 to 5.00	Strongly Agree (Excellent)
4	3.41 to 4.20	Agree (Very Good)
3	2.61 to 3.40	Neutral (Good)
2	1.81 to 2.60	Disagree (Fair)
1	1.00 to 1.80	Strongly Disagree (Poor)

Procedure. The proponents held a demonstration and gave each respondent a user manual (Appendix B). Each respondent then performed assigned tasks in the system, such as searching, borrowing, and returning a book, for about 30 minutes. After that, each one answered the questionnaire. The IT experts also reviewed the system's design and security.

Statistical treatment. The **weighted mean** was used to summarize the ratings, computed as $WM = \Sigma(f \times w) / N$, where f is the number of respondents who gave a rating, w is the rating value, and N is the total number of respondents.

UAT sign-off. After the evaluation, the head librarian signed the acceptance form in Table 4.8, confirming that the system was accepted for use.

Table 4.8. UAT Sign-off Form

Item	Details
System name	LibraTrack: A Web-Based Library Borrowing and Inventory System
Test period	[Start date] to [End date]
Result of UAT	The system meets the users' requirements and is accepted for use.
Accepted by	[Name], Head Librarian Signature: _____ Date: _____
Prepared by	[Names of proponents]

Preview of how results appear in Chapter 5 (Section 5.2). The sample table below shows how the mean scores from the ISO/IEC 25010 evaluation may be presented. The numbers are made up for illustration.

Table 5.x. Mean Scores per Quality Characteristic (Sample Data Only)

Quality Characteristic	Weighted Mean	Verbal Interpretation
Functional Suitability	4.52	Strongly Agree (Excellent)
Performance Efficiency	4.38	Strongly Agree (Excellent)
Compatibility	4.41	Strongly Agree (Excellent)
Usability	4.60	Strongly Agree (Excellent)
Reliability	4.35	Strongly Agree (Excellent)
Security	4.44	Strongly Agree (Excellent)
Maintainability	4.30	Strongly Agree (Excellent)
Portability	4.36	Strongly Agree (Excellent)
Overall Mean	**4.42**	**Strongly Agree (Excellent)**

Common Mistakes to Avoid

- Listing only test cases that pass. Show failed cases and how you fixed them.
- Testing only valid input. Include invalid data and boundary values.
- Test cases that do not match your modules, ERD, and requirements.
- Performance goals with no numbers (write "3 seconds or less", not "fast").
- Putting survey results in Section 4.5. Methods go in Chapter 4; results and discussion go in Chapter 5.
- A questionnaire that does not follow ISO/IEC 25010, or a scale with no mean ranges.
- Ask your adviser which edition of ISO/IEC 25010 your department follows, since the standard was revised in 2023.